

01

MAPPING FLOOD DAMAGE FROM SPACE

AT A GLANCE

Duration: 15 days

Teams: Upto 4 members

Data: free and open satellite,
elevation and map data

Case study: the August 2026
Trishuli flood, Nepal

Evaluation: live, on areas and
dates chosen by the judges on
the day

When roads, bridges and phone lines are gone, rescuers still need to know where the damage is and who is cut off. Build the system that tells them — from space.

Organising Committee

On 26 August 2026, the partial collapse of a high-altitude glacier in northern Nepal triggered an avalanche of ice and rock that sent a flood of water, mud and debris down the Bhote Koshi–Trishuli river corridor. Nepal’s National Disaster Risk Reduction and Management Authority reported 1,342 deaths by 6 September, with almost 4,900 people still missing or out of contact (NDRRMA situation report, as reported by the Copernicus Emergency Management Service).

In disasters like this, ground sensors can be swept away, and roads, bridges and communications fail. Rescuers urgently need to know where the damage is worst and which villages have been cut off. Free satellite data covers every valley of the Himalaya. Radar satellites see through monsoon clouds, and optical satellites show damage in detail. The challenge is turning that raw data into trustworthy answers, quickly enough to matter.



BEFORE

A radar view of a mountain valley. The river shows as a thin dark line; slopes facing the satellite appear bright. (Illustrative image.)



AFTER

The same valley after a debris flood: a wide, rough corridor of mud and rock with dark ponded water. Your system finds this change. (Illustrative.)

YOUR TASK

Build a system that takes **an area** and **a flood date** and uses satellite images to answer three questions for rescue teams.

1. Where did the flood hit? Map flooded and debris-covered areas by comparing images from before and after the event. Use Sentinel-1 radar, which sees through monsoon clouds, and Sentinel-2 optical images when the sky is clear.

2. What was damaged? Overlay buildings, roads and bridges from OpenStreetMap and estimate which ones were hit.

3. Who is cut off? Using the road network, find settlements that no longer have a road connection to the nearest town or hospital.

“Satellites can see the valleys where ground sensors were swept away.”

Why this challenge matters

AI COMPONENT

Choose one of the following.

Flood segmentation model. Train a model to detect water or debris in satellite images using the listed training data, and show how well it works on Himalayan scenes it has never seen.

Situation-report copilot. An assistant that answers rescuers’ questions and writes short situation reports in English and Nepali. Every number must come from your system’s maps, never invented by the language model.

BONUS

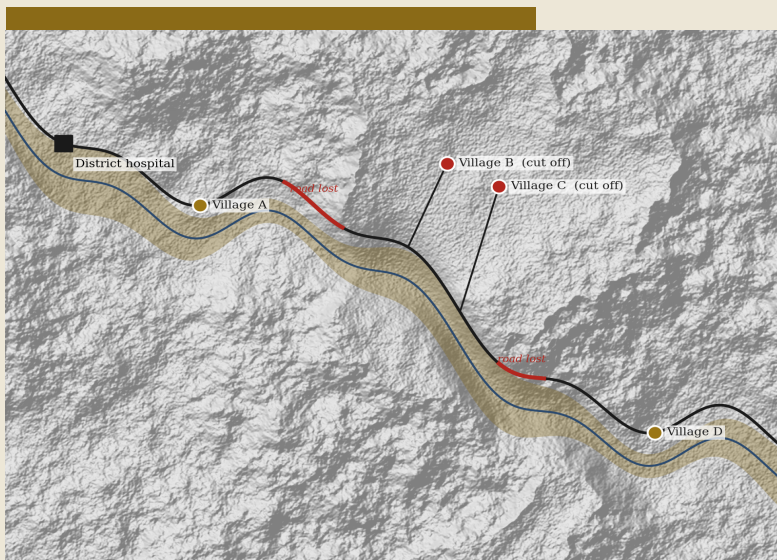
Given any point upstream, trace the flood path down the valley using elevation data and list the settlements along it.

OUTPUT

A map dashboard, plus a one-page situation report generated by the system.

LIMITATIONS

Explain what your system *cannot* do. For example, Sentinel satellites revisit an area only every few days, so they cannot warn of a sudden glacier collapse minutes ahead.



WHO IS CUT OFF?

The flood (gold) has destroyed two stretches of road (red). Villages B and C can no longer reach the district hospital by road, while villages A and D still can.

This is the kind of answer your system should produce automatically, for any valley the judges choose.

Illustrative example, not real data.

DATA RULES

Allowed as inputs: Sentinel-1, Sentinel-2, Copernicus DEM, OpenStreetMap *as it was before the event*, and the listed training datasets.

Not allowed as inputs: Copernicus EMS, UNOSAT or any other published damage maps, and OpenStreetMap edits made after the event. These may only be used to *check* your results. Breaking this rule leads to disqualification.

The system must run end-to-end from raw satellite data for *any* area and date the judges choose.

CASE STUDY

Map the August 2026 Trishuli flood and compare your results with the official Copernicus Emergency Management Service maps for this event (activation **EMSR927**).

Tip: Sentinel-1 radar images of the corridor exist from a few days before and after the event. For change detection, compare images from the *same orbit track* (same time of day, 12 days apart). Images from different tracks view the terrain from different angles and cannot be compared pixel by pixel.

DATASETS AND LICENSES

DATASET	USE	LICENSE AND CONDITIONS
Sentinel-1 and Sentinel-2 dataspace.copernicus.eu	Before/after images	Copernicus free, full and open data policy; any use allowed. Free registration.
OpenStreetMap pre-event snapshot via the ohsome API https://api.ohsome.org/v1	Buildings, roads, bridges	ODbL. Use a snapshot date before 26 Aug 2026 (currently the latest available is 27 July 2026). A modified version of the OSM data itself, if published, must stay under ODbL.
Kuro Siwo (recommended) https://github.com/Orion-AI-Lab/KuroSiwo	Training data	MIT License per the NeurIPS 2024 paper (the repository README also states CC BY). Free use with credit; cite Bountos et al., 2024.
Sen1Floods11 (optional) https://github.com/cloudtostreet/Sen1Floods11	Training data	The original repository has no license file; mirrors list CC BY 4.0. Fine for non-commercial educational use; cite Bonafilia et al., CVPR Workshops 2020.
Copernicus EMS, EMSR927 https://mapping.emergency.copernicus.eu	Checking only	Free of charge. Credit “European Union, Copernicus Emergency Management Service data”.

REQUIRED ATTRIBUTION IN EVERY SUBMISSION

“Contains modified Copernicus Sentinel data 2026.”

“Produced using Copernicus WorldDEM-30 © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018 provided under COPERNICUS by the European Union and ESA; all rights reserved.”

“© OpenStreetMap contributors.” Plus a citation of any training dataset used.

A NOTE ON RESPONSIBILITY

This disaster is recent, and many families are still affected. Treat the topic with respect: do not use images of victims in demos or reports. Systems built here are educational prototypes, not operational tools.

JUDGING

Flood and damage mapping quality	30%
AI component	25%
Cut-off settlement analysis	20%
Report, understanding in Q&A, and honesty about limitations	15%
Dashboard and situation report usability	10%

SUBMISSION CHECKLIST

- GitHub repository with a README explaining how to run the system
- Report (max. 6 pages), including the limitations section
- 3-minute demo video
- Required data attributions and training-dataset citations
- Case-study results compared with the EMSR927 reference maps

Images in this document are illustrative, generated for this problem statement; they are not real satellite data.